

Manifold Theory

PENG

Created: 20260324, Version: 20260627

Contents

1	Set Theory	1
1.1	Relations	1
2	Euclidean Spaces	2
3	Topological Spaces	2

1 Set Theory

1.1 Relations

Definition 1.1. *Equivalence relation $\sim$:*

- *reflexive:* $x \sim x$
- *symmetric:* $x \sim y \Rightarrow y \sim x$
- *transitive:* $x \sim y, y \sim z \Rightarrow x \sim z$

Definition 1.2. *Equivalence class of each $x \in X$:*

$$[x] = \{y \in X : y \sim x\}$$

Definition 1.3. *The collection of all $[x]$ is $X/\sim$*

Example 1.4. *Let $A = \{1, 2, 3\}$, $R = \{(x, y) : x = y\}$ on A*

- *reflexive:* $x = x$
- *symmetric:* $x = y \Rightarrow y = x$
- *transitive:* $x = y, y = z \Rightarrow x = z$

Thus R is an equivalence relation.

Equivalence class:

$$[1] = \{1\}$$

$$[2] = \{2\}$$

$$[3] = \{3\}$$

Example 1.5. *Let $A = \{1, 2, 3\}$, $R = \{(x, y) : x + y \text{ is even}\}$ on A*

- *reflexive:* $x + x \text{ is even}$
- *symmetric:* $x + y \text{ is even} \Rightarrow y + x \text{ is even}$
- *transitive:* $x + y \text{ is even}, y + z \text{ is even} \Rightarrow \begin{cases} x \text{ is even}, y \text{ is even}, z \text{ is even} \\ x \text{ is odd}, y \text{ is odd}, z \text{ is odd} \end{cases} \Rightarrow x + z \text{ is even}$

Thus R is an equivalence relation.

Equivalence class:

$$[1] = \{1, 3\}$$

$$[2] = \{2\}$$

$$[3] = \{1, 3\}$$

Definition 1.6. A partition of X is a collection of disjoint nonempty subsets of X whose union is X .

Example 1.7. $X = \{1, 2, 3\}$

$$P = \{\{2\}, \{1, 3\}\}$$

$$P = \{\{1\}, \{3\}, \{2\}\}$$

$$P = \{\{1, 2, 3\}\}$$

$\{\{1\}, \{2\}, \{3\}\}$: nonempty

$\{\{1\}, \{2\}\}$: union is not X

$\{\{1, 3\}, \{2, 3\}\}$: disjoint

$\{\{1, 3\}, \{2, 4\}\}$: union is not X

Exercise 1.8. Collection of equivalence class $\Rightarrow$ partition

Partition $\Rightarrow$ unique equivalence relation

Proof. $X/\sim \Rightarrow$ partition:

For $\forall x_1, x_2 \in X$:

$$[x_1] = \{y \in X : y \sim x_1\}$$

$$[x_2] = \{y \in X : y \sim x_2\}$$

If $x_1 \not\sim x_2, \forall y \in [x_1] \Rightarrow y \notin [x_2] \Rightarrow$ disjoint nonempty subsets

If $x_1 \sim x_2, \forall y \in [x_1] \Rightarrow y \in [x_2] \Rightarrow [x_1] = [x_2]$ □

Proof. Partition $\Rightarrow \sim$ □

2 Euclidean Spaces

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

Definition 2.1. For vectors:

$$x \cdot y = x^T y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

Definition 2.2. Norm of x :

$$|x| = \sqrt{(x_1)^2 + (x_2)^2 + \cdots + (x_n)^2}$$

Exercise 2.3.

$$\max\{|x_1|, |x_2|, \dots, |x_n|\} \leq |x| \leq \sqrt{n} \max\{|x_1|, |x_2|, \dots, |x_n|\}$$

Proof. Let $x_m = \max\{|x_1|, |x_2|, \dots, |x_n|\}$:

$$\sqrt{(x_m)^2} \leq \sqrt{(x_1)^2 + \cdots + (x_m)^2 + \cdots + (x_n)^2} \leq \sqrt{(x_m)^2 + \cdots + (x_m)^2}$$

□

Definition 2.4. θ between x and y :

3 Topological Spaces

Topology [1]

Four mechanisms (normal, segment, merge, torus) exist in every fluid point: fractal.

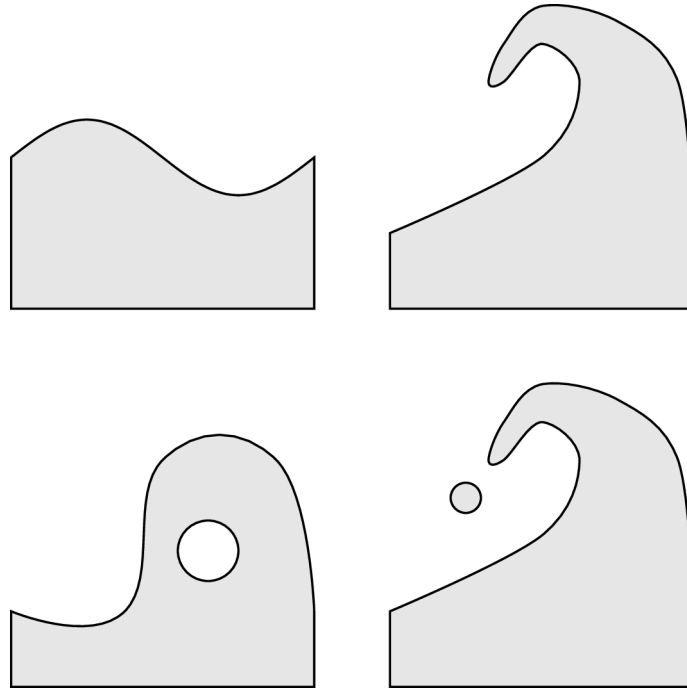


Figure 3.1: Sloshing topology.

References

- [1] J.M. Lee. *Introduction to Topological Manifolds*. Springer New York, 2010. ISBN: 9781441979391.